

# Samaranjan Manjhi

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## EDUCATION

| Degree          | University        | Institute                                     | Year      | CGPI/% |
|-----------------|-------------------|-----------------------------------------------|-----------|--------|
| Graduation      | Mumbai University | Shree LR Tiwari College Of Engineering, Thane | 2019-2023 | 8.56   |
| Intermediate/+2 | HSC               | Thakur College Of Science & Commerce, Mumbai  | 2017-2019 | 74.00  |
| Matriculation   | SSC               | Thakur Shyamnarayan High School, Mumbai       | 2016-2017 | 89.60  |

## PROFESSIONAL EXPERIENCE

**Programmer** August 2024 – Present

*MicroWorld Software Services Pvt Ltd* *Andheri, Mumbai*

- Developed a macOS process monitoring tool and file integrity monitoring system, enabling real-time tracking of process executions and file changes, with alerting and event logging for enhanced security.
- Built a cross-platform cloud backup and monitoring tool and content scanning solution, ensuring data consistency, preventing data loss, and identifying sensitive information across multiple file formats.

**Trainee Programmer** August 2023 – August 2024

*MicroWorld Software Services Pvt Ltd* *Andheri, Mumbai*

- Developed Linux kernel module with system call hooks for file operation monitoring and ransomware protection.
- Built vulnerability checker application with Qt GUI integrating & SQLite for security scanning and management.

## KEY PROJECTS

**ProcMon - Process Monitoring Tool** | *C/C++, libmagic, bsm* July 2025 - Sept 2025

- Developed a lightweight macOS process monitoring tool that logs new executions (e.g., scp, rsync, vim, file copy) with command-line arguments and process hierarchies in real time using system calls, without relying on Endpoint Security or external libraries.

**File Integrity Monitoring System** | *C/C++, OpenSSL, sqlite3* June 2025 - Aug 2025

- Developed a macOS File Integrity Monitoring (FIM) tool that loads directory watch lists from config, stores metadata (permissions, MD5, size, timestamps) in SQLite, and continuously monitors for real-time changes (permission/content/deletion) with alerting and event logging for security auditing.

**cBackup - Cloud Backup & Monitoring Tool** | *C/C++, OpenSSL, cURL* Feb 2025 - May 2025

- Developed cBackup (Desktop EDR Edition), a cross-platform backup and monitoring tool for macOS/Linux that tracks file transfers (USB, scp, rsync, curl, email), auto-backs up files locally (`_attach/_usb`), and synchronizes them across multiple cloud services with 10-minute cycles ensuring consistency and data loss prevention.

**Content Scanner** | *C/C++, Leptonica, Tesseract, Poppler* Sept 2024 - Dec 2024

- Developed a data scanning tool that identifies confidential data (PAN, emails, etc.) across 12+ file types (PDF, images, XML, DOCX, archives) using regex patterns; deployable as both a shared library and a CLI for versatile integration.

**System Call Interception Module** | *C, Linux Kernel, Hooking* Mar 2024 - Aug 2024

- Developed a Linux kernel module to intercept system calls and implement custom hooks for monitoring file operations by extension (creation, renaming, moving), strengthening file system protection and enhancing security against ransomware and malicious activities.

**Vulnerability Checker Application** | *C/C++, Qt, Boost, cURL, OpenSSL, cJSON* Oct 2023 - Jan 2024

- Developed a secure vulnerability checker application by integrating front-end and back-end technologies, leveraging cURL for ZIP downloads, XML parsing, and efficient SQLite storage to ensure seamless functionality, organized scanning, and reliable data management.

## TECHNICAL SKILLS

**Languages:** C/C++

**Frameworks:** Qt

**Operating System:** Linux, macOS

**Developer Tools:** Vim, Xcode, Git, SQLite

**Libraries:** boost, cJSON, curl, leptonica, tesseract, poppler, libmagic, libarchive